

Lorenz eqns.

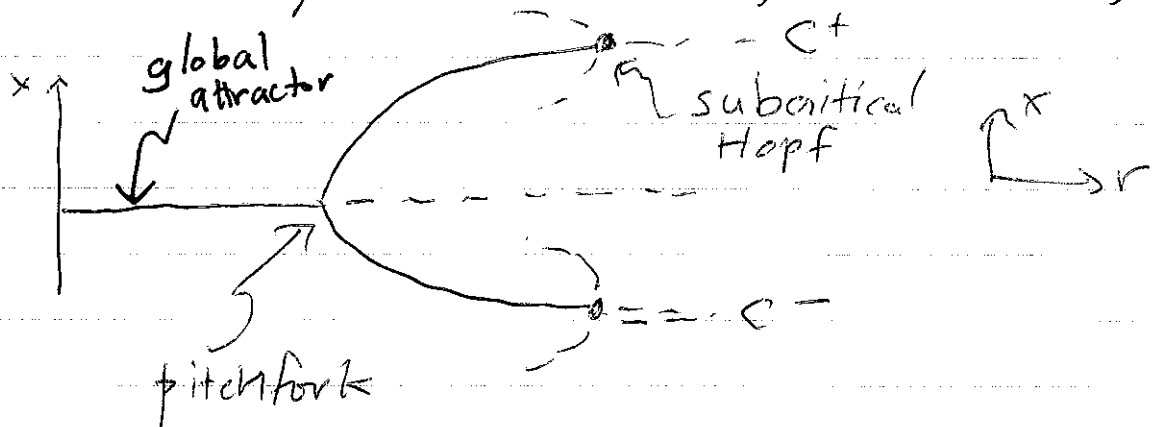
$$\left. \begin{aligned} \dot{x} &= \sigma(y-x) \\ \dot{y} &= rx - y - xz \\ \dot{z} &= xy - bz \end{aligned} \right\} \quad \sigma, r, b > 0$$

Fixed-pts

$$\bullet \quad x=y=z=0$$

$$\bullet \quad x=y, \quad z = x^2/b = (r-1) > 0$$

$$C^{\pm}: \quad x^* = y^* = \pm \sqrt{b(r-1)}; \quad z^* = (r-1), \quad r > 1$$



$r < 1$: $x=y=z=0$ is global attractor

Proof : $V = \frac{1}{\sigma} x^2 + y^2 + z^2$ is a Lyapunov function for $x=y=z=0$

$$V > 0 \quad \text{for} \quad (x, y, z) \neq (0, 0, 0)$$

$$V = 0 \quad \text{for} \quad (x, y, z) = (0, 0, 0)$$

$$\frac{dV}{dt} = 2 \left(\frac{1}{\sigma} x \dot{x} + y \dot{y} + z \dot{z} \right)$$

$$= 2 \left(xy - x^2 + rxy - y^2 - \cancel{xy}z + \cancel{yz} - bz^2 \right)$$

$$= -2 \left[\underbrace{x^2 - (r+1)xy + y^2}_{\left(x - \frac{(r+1)}{2}y\right)^2 - \frac{(r+1)^2}{4}y^2} + bz^2 \right]$$

$$\left(x - \frac{(r+1)}{2}y\right)^2 - \frac{(r+1)^2}{4}y^2 + y^2 + bz^2$$

$$= -2 \left[\left(x - \frac{(r+1)}{2}y\right)^2 + \left(1 - \frac{(r+1)^2}{4}\right)y^2 + bz^2 \right]$$

$$< 0 \quad \text{for } (x, y, z) \neq (0, 0, 0)$$

$$\text{provided } 1 - \frac{(r+1)^2}{4} > 0 \Rightarrow 1 > \frac{r+1}{2} \Rightarrow r < 1$$

$r=1$ is pitchfork bifurcation pt. :

$$DF(0,0,0) = \begin{bmatrix} -\sigma & +\sigma & 0 \\ r & -1 & 0 \\ 0 & 0 & -b \end{bmatrix}$$

eigenvalues: $\mu = -b < 0$

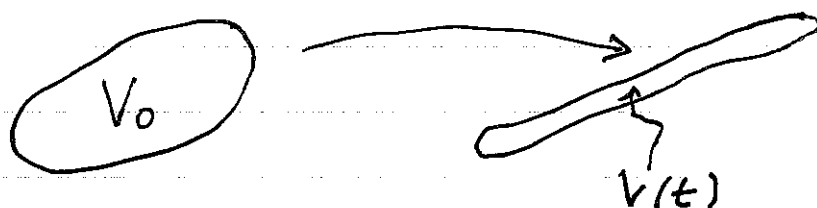
$$\begin{bmatrix} -\sigma & +\sigma \\ r & -1 \end{bmatrix} \Rightarrow$$

$$\tau = -(\sigma + 1) < 0$$

$$\Delta = \sigma - r\sigma$$

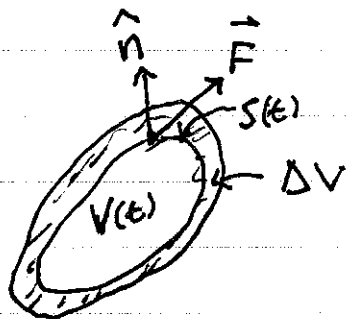
$$\Delta = 0 \text{ at } r = 1$$

Lorenz eqns. : Example of a dissipative system - phase space volumes contract exponentially in time:



consider small unit of time Δt where $V(t) \rightarrow V(t+\Delta t)$

$$V(t+\Delta t) = V(t) + \iint_{S(t)} \vec{F} \cdot \hat{n} \Delta t \, dS$$



$$\lim_{\Delta t \rightarrow 0} \frac{\Delta V}{\Delta t} = \frac{dV}{dt} = \iint_{S(t)} \vec{F} \cdot \hat{n} \, dS$$

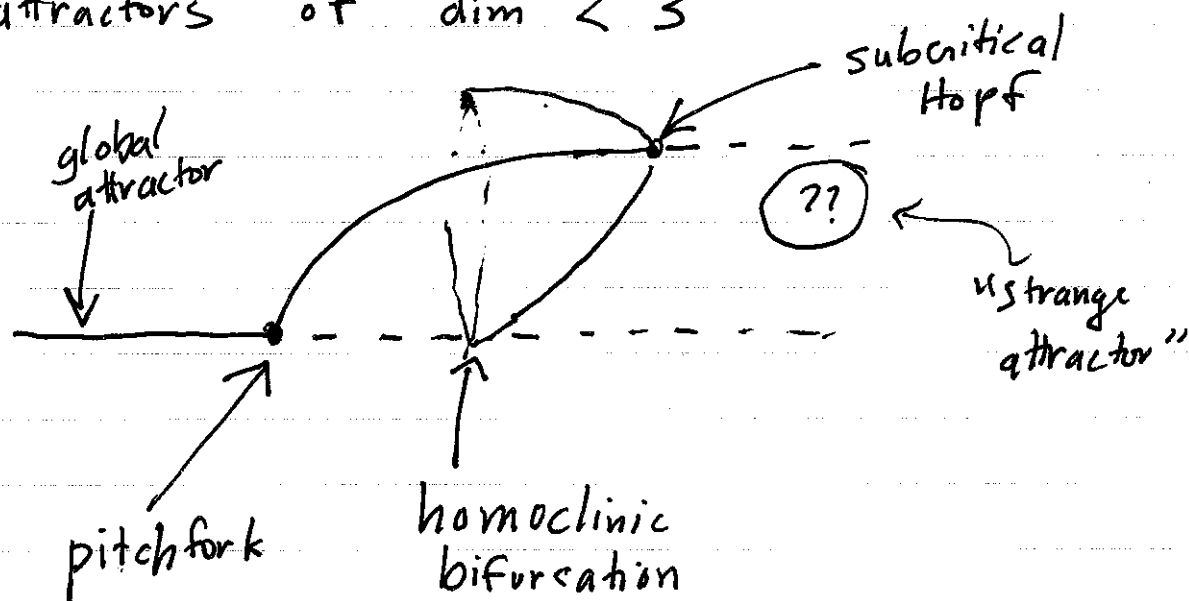
$$\frac{dV}{dt} = \iiint_{V(t)} \nabla \cdot \vec{F} \, dV \quad \text{by divergence theorem}$$

$$\begin{aligned} \nabla \cdot \vec{F} &= \frac{\partial}{\partial x} [\sigma(y-x)] + \frac{\partial}{\partial y} [rx-y-xz] + \frac{\partial}{\partial z} [xy-bz] \\ &= -\sigma - 1 - b = -(\sigma + 1 + b) < 0 \end{aligned}$$

$$\frac{dV}{dt} = -(\sigma + 1 + b) \iint dV = -(\sigma + 1 + b)V, \quad V(0) = V_0$$

$$\Rightarrow V(t) = e^{-(\sigma + 1 + b)t} V_0 \rightarrow 0 \text{ as } t \rightarrow \infty$$

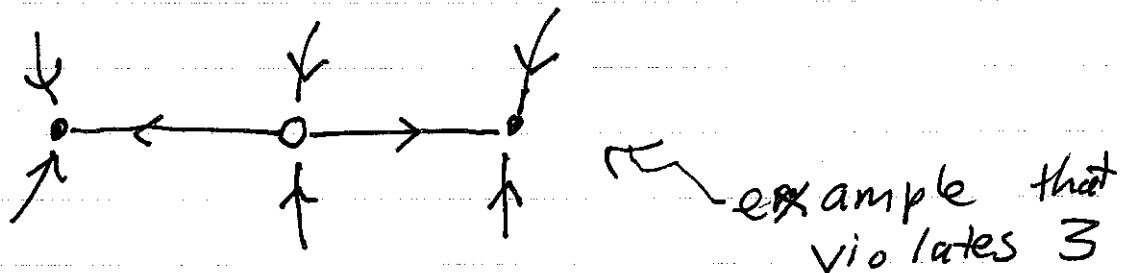
long-term dynamics is associated with attractors of $\dim < 3$



Attractor: closed set A with properties

1. A is an invariant set: any trajectory $x(t)$ that starts in A stays in A for all future time
2. A attracts an open set of initial conditions: there is a U containing A s.t. if $x(0) \in U$ then the distance between $x(t)$ & A goes to 0 as $t \rightarrow \infty$. Largest such U = "basin of attraction" of A .

3. A is minimal : there is no proper subset of A that satisfies 1 & 2.



Lorenz's "strange attractor" is strange in ~~that~~ its fractal nature.

- volume collapses to zero so $\dim(A) < 3$
- for chaos we need $\dim(A) > 2$

Lorenz described its fractal nature - orbits move as if on sheets, but the trajectories cannot cross, so there must be an infinite number of them.

Some notions of dimension:

Box dimension: count the number of "boxes" needed to cover a set as a function of the size of the boxes. Take limit as box size $\rightarrow 0$.